

CE 525 : Advance Transportation Engineering

Course Code : CE 525

Course Name : Advance Transportation Engineering

L-P-T-C: 3-0-1-4

Intended for: B. Tech (final year); MS (Research); PhD

Prerequisites: NA

Mutual Exclusion: NA

Approval : 57th BoA

Course Contents

- Section 1: **Pavement Materials and Design**

- **Pavement Materials:** Mechanical characteristics for the pavement materials for design sections. Deformation characteristics of unbound layers (CBR, resilient modulus, modulus of subgrade reaction), functional properties (permeability) etc. 14 hrs)
- **Design of bituminous & Concrete mixes:** Requirement of bitumen mixes, design of bituminous mixes as per Marshall Stability & flow method, parametric evaluation of bituminous mixes, IRC & MoRTH recommendations for the design mix of various layers of flexible pavements. [IRC 44] [6 hrs)
- **Design of Flexible Pavements:** Stresses in flexible pavements, theories of stress distribution, Boussinesq's Elastic theory, Burmister's theory, considerations for flexible pavement design, IRC method & other countries method for the design of flexible pavements, AASHTO method of Pavement design [IRC: 37 2018] [8 hrs]
- **Design of Rigid Pavements:** Stresses in rigid pavements, Westergaard method of rigid pavement design, IRC method of rigid pavement design for plain dowel jointed slabs, design of joints and load transfer devices; design of tie bars, joint fillers and sealers, design of continuously reinforced concrete pavements, design of thin & ultra-thin white toppings as overlay. [IRC 58:2015] [8 hrs)

- Section 2: **Highway Planning and Design**

- **Traffic Planning& control:** Fundamental principles of Traffic Flow, Traffic flow Elements, Flow Density Relationships, Traffic signs, Road markings, traffic signals, type's i.e. simultaneous system, alternate system, simple progressive system and flexible progressive system, general principles of signal design, Roadway delineations, object markers, guard rails, Barriers. [6 hrs]
- **Highway capacity& Intersection design:** PCU, Level of service concepts, factors affecting capacity, capacity of urban highways, capacity of rotary intersection, Design of intersection, grade separated intersection, Need for rotary intersection, principles of design, design of rotary intersection. [6 hrs]

Lab Sessions: (8 Hours)

1. Testing and Characterization of Pavement Materials o Laboratory Work: I. Plate bearing test.
2. Stability and Flow value test of bituminous mix as per Marshall Criteria.
3. Evaluation of pavement by Benkelman beam.

4. Evaluation of pavement roughness by Bump Integrator.
5. Introduction to highway engineering softwares(HEADS, MX Road and HDM4)
6. To conduct Pavement Deterioration tests.
- o Design and Analysis of Pavement

Text books:

1. Khanna S.K. and C.E.G. Justo, **Highway Engineering**, Nemchand Bros, 2002.
2. Kadyali L. R., **Highway Engineering**, Nem Chand & Brothers, 2002.

References:

1. Sharma & Sharma, **Principle and Practice of Highway Engineering**, Asia Publishing House, New Delhi (1980).
2. **Road Development plan for India** (1981-2001), IRC, New Delhi, (1984).
3. Rao G. V., **Transportation Engineering**, Tata McGraw Hill Publisher, New Delhi
4. **Highway Materials**, H.M.S.O. (London).
5. Yoder E. J., **Principles of Pavement Design**, John Wiley & Sons, 1975.
6. Haas R.C.G., Hudson W. Ronald., Zaniwski John P., **Modem Pavement Management**, Krieger Publishing Company, 1994.
7. Susan Brown, **Pavement Managaement Systems**, Transportation Research Board, 1993.